ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SEMESTER PROJECT SPRING 2026

BACHELOR IN MATHEMATICS

Minimax Goodness-of-fit testing in High-Dimensional models

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Chapter 1

Introduction

One of the most fundamental questions in statistics is deceptively simple: given a dataset, does it come from a distribution we have in mind, or not? This is the goodness-of-fit testing problem, and it has been studied for over a century. Classical tests such as Kolmogorov–Smirnov or Pearson's χ^2 were designed for simple, low-dimensional settings. But as modern scientific applications generate data of increasing complexity and dimension, a natural question arises: how many observations do we actually need to answer this question reliably, and can we do so efficiently?

This project addresses these questions in a nonparametric minimax framework, where one seeks to characterize the fundamental statistical limits of the problem, that is, the minimum number of samples required by *any* test, regardless of its construction. The setting we focus on is the **Gaussian sequence model** with mean vector constrained to a Sobolev ellipsoid, a canonical model in nonparametric statistics that captures the essential difficulty of signal detection in function spaces. In this model, the unknown signal is represented as an infinite sequence of Gaussian observations, and the smoothness parameter s of the Sobolev ellipsoid controls how fast the signal's frequency components decay. This makes it possible to study, in a precise and clean way, how the regularity of the signal affects the difficulty of testing.

Two approaches to this problem are developed in this project. The first, covered in Chapter 3, is the classical approach based on χ^2-type tests, pioneered by Ingster and Suslina [2012]. These tests aggregate a weighted sum of squared observations and reject the null hypothesis when this sum is large. The minimax theory for such tests is well understood, and they are known to be optimal in the Gaussian sequence model. The second approach, covered in Chapter 4, is more recent and comes from a surprising direction: machine learning. The **Classifier-Accuracy Test** (CAT), introduced by Gerber et al. [2023], replaces the classical test statistic by a binary classifier trained to distinguish samples from the null and alternative distributions. Rather than computing a closed-form statistic, the classifier implicitly learns a *separating set* (i.e. a region of the sample space where the two distributions differ most) and the test is based on how many observations fall in this region.

At first glance, restricting the classifier to a binary output might seem wasteful: the Neyman–Pearson lemma tells us that the optimal test should aggregate the full log-likelihood ratio, not just threshold it. However, one of the main findings of Gerber et al. [2023] is that this intuition is misleading: a well-chosen binary separating set is sufficient to achieve minimax optimality, at least in the sense of sample complexity. This result bridges two seemingly distant fields, nonparametric statistics and machine learning and suggests that classifier-based testing is not merely a practical heuristic, but a theoretically grounded procedure.

The project is organized as follows. Chapter 2 introduces the general minimax framework for hypothesis testing, including the Bayesian and minimax approaches and the connection between them. Chapter 3 specializes to the Gaussian sequence model and develops the classical theory of minimax distinguishability and χ^2-type tests. Chapter 4 introduces the CAT framework, establishes the key lemmas needed to analyze the performance of the separating set, and proves the minimax lower bound on the sample complexity.

Chapter 2

Introduction to Minimax Hypothesis testing

2.1 Goodness-of-Fit testing : Definitions and criteria

Let $(\mathcal{X}, \mathcal{A})$ be a measurable space and let $\mathcal{P}$ be a family of probability measures on it. Given two disjoint subfamilies $\mathcal{P}_0, \mathcal{P}_1 \subseteq \mathcal{P}$ with $\mathcal{P}_0 \neq \emptyset$, the *goodness-of-fit testing problem* consists in deciding, from a single observation $X \sim P$, between

$$H_0 : P \in \mathcal{P}_0 \qquad \text{versus} \qquad H_1 : P \in \mathcal{P}_1.$$

The hypothesis H_0 is called *simple* if $\mathcal{P}_0 = \{P_0\}$, and *composite* otherwise.

Definition 2.1 (Test). *A (randomized)* test *is a measurable function $\psi : \mathcal{X} \to [0,1]$: upon observing $X = x$, the null hypothesis H_0 is rejected with probability $\psi(x)$ and accepted with probability $1 - \psi(x)$. If ψ takes values only in $\{0,1\}$, the test is said to be* nonrandomized*, and the set $\mathcal{X}_{\mathrm{cr}} = \{x \in \mathcal{X} : \psi(x) = 1\}$ is its* critical region*. We denote by Ψ the set of all tests.*

Two error functions are naturally associated to any test $\psi \in \Psi$:

- the *type I error* (rejecting H_0 when it holds) :

$$\alpha(\psi, P) = \mathbb{E}_P[\psi], \qquad P \in \mathcal{P}_0;$$

- the *type II error* (accepting H_0 when it does not hold) :

$$\beta(\psi, P) = \mathbb{E}_P[1 - \psi], \qquad P \in \mathcal{P}_1.$$

For composite hypotheses, the overall quality of a test is measured by the worst-case errors

$$\alpha(\psi, \mathcal{P}_0) = \sup_{P \in \mathcal{P}_0} \alpha(\psi, P), \qquad \beta(\psi, \mathcal{P}_1) = \sup_{P \in \mathcal{P}_1} \beta(\psi, P).$$

Since reducing one error typically increases the other, two classical strategies are used to select a test:

- **Neyman–Pearson approach.** Fix a significance level $\alpha \in (0,1)$, restrict attention to the set

$$\Psi_\alpha = \Psi_{\alpha, \mathcal{P}_0} = \big\{\psi \in \Psi : \alpha(\psi, \mathcal{P}_0) \leq \alpha\big\},$$

  and seek to minimize the type II error over Ψ_α.

- **Combined-error approach.** For a weight $t \geq 0$, minimize the weighted sum $t\,\alpha(\psi, \mathcal{P}_0) + \beta(\psi, \mathcal{P}_1)$ over all tests $\psi \in \Psi$. The case $t = 1$ yields the *total error* $\gamma(\psi) = \alpha(\psi, \mathcal{P}_0) + \beta(\psi, \mathcal{P}_1)$.

When $\mathcal{P}_0$ or $\mathcal{P}_1$ is composite and of dimension greater than one, the set of admissible tests is too large to admit general optimality recommendations. A natural way around this is to characterize tests not through their pointwise error functions $\alpha(\psi, P), \beta(\psi, P)$, but through monotone functionals $F_1(\psi), F_2(\psi)$ aggregating the errors over $\mathcal{P}_0$ and $\mathcal{P}_1$. The two canonical choices, either *averaging* (Bayesian approach, Section 2.2) or *taking the supremum* (minimax approach, Section 2.3). These two approaches are developed in the next two sections.

2.2 Bayesian approach for goodness-of-fit testing

One can note that since the set of admissible test is too wide, there are no general recommendations as to how to select optimal tests when the alternative is composite and of dimension more than one. A way out of this situation is to characterize the test, not in the usual way by using error functions $\alpha(\psi, P), \beta(\psi, Q)$ for some $P \in \mathcal{P}_0, Q \in \mathcal{P}_1$, but by some monotone functional $F_1(\psi), F_2(\psi)$

2.2.1 General setting

Fix a Borel σ-algebra on the space $\mathcal{P}$ of probability measures (with respect to the total variation distance) such that $\mathcal{P}_0, \mathcal{P}_1$ are Borel subsets. Moreover, fix two priors π_0, π_1 with supports in $\mathcal{P}_0, \mathcal{P}_1$, respectively. Now, define two functionals F_1, F_2 as the average errors with respect to each prior :

$$\begin{aligned} F_1(\psi) &:= \alpha(\psi, \pi_0) = \mathbb{E}_{\pi_0}\alpha(\psi, P) = \int \alpha(\psi, P)d\pi_0(P) \\ F_2(\psi) &:= \beta(\psi, \pi_1) = \mathbb{E}_{\pi_1}\beta(\psi, P) = \int \beta(\psi, P)d\pi_1(P) \end{aligned} \tag{2.1}$$

Analogously to what has been done with the standard approach, given $\alpha \in (0, 1)$ we define the set :

$$\Psi_{\alpha,\pi_0} := \{\psi \in \Psi : \alpha(\psi, \pi_0) \leq \alpha\} \tag{2.2}$$

and the following functions :

$$\begin{aligned} \gamma_t(\psi; \pi_0, \pi_1) &:= t\alpha(\psi, \pi_0) + \beta(\psi, \pi_1) \\ \gamma(t; \pi_0, \pi_1) &:= \inf_{\psi \in \Psi} \gamma_t(\psi; \pi_0, \pi_1) \end{aligned} \tag{2.3}$$

$$\beta(\alpha; \pi_0, \pi_1) := \inf_{\psi \in \Psi_{\alpha,\pi_0}} \beta(\psi, \pi_1) \tag{2.4}$$

The tests $\psi^{(t)}$ or ψ_α are called Bayesian if they provide an infimum eq. (2.3) or in eq. (2.4).

Definition 2.2 (Mixture of a prior). *Given a prior π, one can define its mixture as the measure P_π on $(\mathcal{X}, \mathcal{A})$ given by :*

$$\forall A \in \mathcal{A} : \mathbb{P}_\pi(A) = \mathbb{E}_\pi P(A) = \int_{\mathcal{P}} P(A)d\pi(P)$$

Given all these definitions, the Bayesian hypothesis testing is reduced to testing of the simple hypotheses with null hypothesis $P = P_{\pi_0}$ and alternative $P = P_{\pi_1}$

Remark. *Note that when we have a parametrization which corresponds to an injective map $\Theta \to \mathcal{P}$, we set a Borel σ-algebra $\mathcal{T}$ on Θ (with respect to some topology) that makes the map measurable, one can assume that the priors π_0, π_1 are defined on $(\Theta, \mathcal{T})$. Hence, in our specific context, one can assume that π_0, π_1 are measures on $(\mathbb{R}^n, \mathcal{B}^n)$ in the finite-dimensional case, or on $(\ell^2, \mathcal{B})$ in the infinite-dimensional case (Here $\mathcal{B}$ denotes a Borel σ-algebra on ℓ^2).*

Add details + define the following measure

$$r(dv) = C^{-1}\pi(dv)e^{-\frac{|v|^2}{2}}, C = \int e^{-\frac{|v|^2}{2}}\pi(dv) = L_\pi(0) \tag{2.5}$$

Proposition 2.1. *Let π_0, π_1 be two priors. Then the following equivalence holds :*

$$P_{\pi_0} = P_{\pi_1} \Leftrightarrow \pi_0 = \pi_1$$

Proof. Clearly, if $\pi_0 = \pi_1$, then $P_{\pi_0} = P_{\pi_1}$. Now, assume that $P_{\pi_0} = P_{\pi_1}$ then $L_{\pi_0} = L_{\pi_1}$ almost surely. First, consider the finite-dimensional case $x \in \mathbb{R}^n$. Let μ_1, μ_2 be the probability measures defined in eq. (2.5) for the priors π_0, π_1. By the **previous** proposition, the functions L_{π_0}, L_{π_1} are analytical, this implies that the equality $L_{\pi_0} = L_{\pi_1}$ holds true in the whole space $\mathbb{C}^n$. In particular, we get the equality for $z = ix, x \in \mathbb{R}^n$ which corresponds to the equality of characteristic functions of the probability measures μ_0, μ_1. Therefore $\mu_0 = \mu_1$ which yields $\pi_0 = \pi_1$. In the infinite-dimensional case, the proof above yields equality for all finite dimensional marginals, which implies $\pi_0 = \pi_1$. □

2.3 Minimax approach in Hypothesis testing

2.3.1 General setting

Instead of considering the average error, we consider maximum error, in this section, define :

$$F_1(\psi) = \alpha(\psi, \mathcal{P}_0) = \sup_{P \in \mathcal{P}_0} \alpha(\psi, P) \tag{2.6}$$

$$F_2(\psi) = \beta(\psi, \mathcal{P}_1) = \sup_{P \in \mathcal{P}_1} \beta(\psi, P) \tag{2.7}$$

Moreover, similarly to eq. (2.2), given $\alpha \in (0, 1)$ we define :

$$\Psi_\alpha = \Psi_{\alpha,\mathcal{P}_0} = \big\{\psi \in \Psi : \alpha(\psi, \mathcal{P}_0) \leq \alpha\big\} \tag{2.8}$$

One can also define the following quantities :

$$\gamma_t(\psi, \mathcal{P}_0, \mathcal{P}_1) = t\alpha(\psi, \mathcal{P}_0) + \beta(\psi, \mathcal{P}_1)$$

$$\gamma(t, \mathcal{P}_0, \mathcal{P}_1) = \inf_{\psi \in \Psi} \gamma_t(\psi) \tag{2.9}$$

$$\beta(\alpha, \mathcal{P}_0, \mathcal{P}_1) = \inf_{\psi \in \Psi_{\alpha,\mathcal{P}_0}} \beta(\psi, \mathcal{P}_0, \mathcal{P}_1) \tag{2.10}$$

The tests $\psi^{(t)}$ or ψ_α are called *minimax* if they provide infimums for eq. (2.9) or eq. (2.10).

One can observe the following convexity lemma :

Lemma 2.1. *Ingster and Suslina [2012, Lemma 2.3] The following properties hold true :*

1. *The functions $\alpha(\psi, \mathcal{P}_0), \beta(\psi, \mathcal{P}_1), \gamma_t(\psi, \mathcal{P}_0, \mathcal{P}_1)$ are convex in $\psi \in \Psi$*
2. *The function $\alpha \mapsto \beta(\alpha)$ is convex nonincreasing continuous for $\alpha \in [0, 1]$, and the function $t \mapsto \gamma(t)$ is concave nondecreasing continuous in $t \in [0, \infty)$. These two functions are connected by the relations :*

$$\gamma(t) = \inf_{\alpha \in [0,1]} \big(t\alpha + \beta(\alpha)\big) \text{ and } \beta(\alpha) = \sup_{t \in [0,\infty)} \big(\gamma(t) - t\alpha\big) \tag{2.11}$$

2.3.2 Connection with Bayesian approach

Now that the settings of both Minimax and Bayesian approaches were explored, we will see that there exists a close relation between these two. All the following statements go back to Wald [1950]

Definition 2.3 (L_1 distance between two families of measures)**.** *Let A, B be two families of measures, we define the L_1 distance between A and B as follows :*

$$d(A, B) := \inf_{P \in A, Q \in B} TV(P, Q)$$

Moreover, we define $[A]$ to be the convex hull of A.

Theorem 2.1. *Ingster and Suslina [2012, Theorem 2.1]*

1. *The following inequalities hold true :*

$$\gamma(t) \geq \sup\big\{\gamma(t, P_0, P_1) : P_0 \in [\mathcal{P}_0], P_1 \in [\mathcal{P}_1]\big\} \tag{2.12}$$

$$\beta(\alpha) \geq \sup\big\{\beta(\alpha, P_0, P_1) : P_0 \in [\mathcal{P}_0], P_1 \in [\mathcal{P}_1]\big\} \tag{2.13}$$

   In particular, $\gamma = \gamma(1) \geq 1 - \frac{1}{2}d([\mathcal{P}_0], [\mathcal{P}_1])$.

2. *If $\mathcal{P}_0, \mathcal{P}_1$ are dominated families of measures on $(\mathcal{X}, \mathcal{A})$, and the $\sigma-$algebra $\mathcal{A}$ is countably generated, then for any $t > 0, \alpha \in (0, 1)$ the infimas in 2.12 and 2.13 are attained, so the inequalities become equalities.*
3. *If $\mathcal{P}_0, \mathcal{P}_1$ are compact with respect to the total variation distance. Then, there exists pairs of priors $\pi_{0,t}, \pi_{1,t}$ and $\pi_{0,\alpha}, \pi_{1,\alpha}$ such that the minimax tests $\psi^{(t)}, \psi_\alpha$ are optimal tests for testing the simple hypothesis $P_{\pi_{0,t}}$ or $P_{\pi_{0,\alpha}}$ against the simple alternative $P_{\pi_{1,t}}$, or $P_{\pi_{1,\alpha}}$. The priors $\pi_{0,t}, \pi_{1,t}$ and $\pi_{0,\alpha}, \pi_{1,\alpha}$ are called least favourable.*

We will use the previous inequalities 2.12,2.13 in order to construct some lower bounds for the quantities $\gamma(t), \beta(\alpha), \gamma$. For example, let $\pi_{0,n}, \pi_{1,n}$ be sequences of priors such that :

$$\pi_{0,n}(\mathcal{P}_0) = \pi_{1,n}(\mathcal{P}_1) = 1 \tag{2.14}$$

Then, one can rewrite the inequalities as :

$$\gamma(t) \geq \limsup_{n\to\infty} \gamma(t; P_{\pi_{0,n}}, P_{\pi_{0,n}}) \tag{2.15}$$

$$\beta(\alpha) \geq \limsup_{n\to\infty} \beta(\alpha; P_{\pi_{0,n}}, P_{\pi_{1,n}}) \tag{2.16}$$

$$\gamma \geq 1 - \frac{1}{2} \liminf_{n\to\infty} d(P_{\pi_{0,n}}, P_{\pi_{1,n}}) \tag{2.17}$$

One can see that this can be also true if $\pi_{0,n}(\mathcal{P}_0) \to 1$ and $\pi_{1,n}(\mathcal{P}_1) \to 1$

Proposition 2.2. *Let $\pi_{0,n}, \pi_{1,n}$ be sequences of priors such that :*

$$\lim_{n\to\infty} \pi_{0,n}(\mathcal{P}_0) = 1, \lim_{n\to\infty} \pi_{1,n}(\mathcal{P}_1) = 1$$

Then, the inequalities 2.15, 2.16 and 2.17 hold true.

Proof. Let us consider conditional priors

$$\tilde{\pi}_{0,n} = \pi_{0,n;\mathcal{P}_0}, \qquad \tilde{\pi}_{1,n} = \pi_{1,n;\mathcal{P}_1},$$

defined by

$$\pi_B(A) = \pi(A \cap B)/\pi(B); \qquad \pi_B(B) = 1,$$

and

$$\begin{aligned} |\pi_B - \pi|_1 &= \int_B |\pi_B(dP) - \pi(dP)| + \pi(\bar{B}) = \int_B \frac{1-\pi(B)}{\pi(B)} \pi(dP) + \pi(\bar{B}) \\ &= \frac{1-\pi(B)}{\pi(B)} \pi(B) + \pi(\bar{B}) = 2\big(1 - \pi(B)\big), \end{aligned} \tag{2.18}$$

where $\bar{B}$ is the complement of B. By continuity $\gamma(t; \mathcal{P}_0, \mathcal{P}_1)$, $\beta(\alpha; \mathcal{P}_0, \mathcal{P}_1)$ on $\mathcal{P}_0, \mathcal{P}_1$ (see Proposition 2.1), it suffices to check that

$$|P_{\tilde{\pi}_{l,n}} - P_{\pi_{l,n}}|_1 \to 0, \qquad l = 0, 1.$$

The last relation follows from Add eq 2.32:

$$|P_{\tilde{\pi}_{l,n}} - P_{\pi_{l,n}}|_1 \ \leq\ |\tilde{\pi}_{l,n} - \pi_{l,n}|_1 \ =\ 2\big(1 - \pi_{l,n}(\mathcal{P}_l)\big) \ \to\ 0, \qquad l = 0, 1.$$

□

Theorem 2.2. *Ingster and Suslina [2012, Theorem 2.3]*
Let π_0, π_1 be priors, and let $S_0 \subseteq \mathcal{P}_0, S_1 \subseteq \mathcal{P}_1$ be measurable sets such that $\pi_0(S_0) = \pi_1(S_1) = 1$. Moreover, assume that the Bayesian tests ψ^t or ψ_α satisfy for $P_0 \in \mathcal{P}_0, P_1 \in \mathcal{P}_1$:

$$\alpha(\psi_\alpha, \mathcal{P}_0) = \alpha(\psi_\alpha, P_0)\ ,\ \beta(\psi_\alpha, \mathcal{P}_1) = \beta(\psi_\alpha, \mathcal{P}_1) \text{ or}$$
$$\gamma_t(\psi^{(t)}; \mathcal{P}_0, \mathcal{P}_1) = \psi(\psi^{(t)}; P_0, P_1)$$

Then, (π_0, π_1) is a pair of least favourable priors under the criteria $\gamma(t)$ or $\beta(\alpha)$, and the tests $\psi^{(t)}$ or ψ_α are minimax.

Proof. Consider the criterion $\beta(\alpha)$ (considerations for the criterion $\gamma(t)$ are analogous). It is clear that $\psi_\alpha \in \Psi_\alpha$, $\beta(\psi_\alpha, \pi_1) = \beta(\psi_\alpha, \mathcal{P}_1)$. Let $\psi' \in \Psi_\alpha$. Then, since the average value is no larger than the maximum value, one has $\alpha(\psi', \pi_0) \leq \alpha(\psi', \mathcal{P}_0) \leq \alpha$ and, since the average value is no smaller than the minimum value, one has

$$\beta(\psi', \mathcal{P}_1) \ \geq\ \beta(\psi', \pi_1) \ \geq\ \beta(\alpha, \pi_1) \ =\ \beta(\psi_\alpha, \pi_1) \ =\ \beta(\psi_\alpha, \mathcal{P}_1).$$

□

In particular, this theorem helps us to obtain the minimaxity of χ^2 tests. However, it is impossible to extend this example to the infinite dimensional case

Chapter 3

Minimax distinguishability in the Gaussian sequence model

In what follows, let $(\mathcal{X}, \mathcal{A})$ be a measurable space a consider a family $\left(P_\theta\right)_{\theta \in \Theta}$ of probability measures on the latter measurable space. Let us have an observation $X \in \mathcal{X}$ generated by P_θ for some unknown parameter $\theta \in \Theta$.

3.1 Gaussian sequence model

Definition 3.1. *We observe a sequence $X = (X_i)_{i \in I} \in \ell^2$ which is of the following form :*

$$X = v + \eta \tag{3.1}$$

Here, $v \in \ell^2$ is the (unknown) mean vector, $\eta_i \overset{i.i.d}{\sim} \mathcal{N}(0,1)$

We denote by P_v the distribution of X when the signal is v, and by P_0 the distribution of X when $v = 0$. Naturally, one can define the following sets :

$$\mathcal{P}_n^G := \left\{P_v, v \in \mathbb{R}^n\right\} \text{ and } \mathcal{P}_\infty^G := \left\{P_v, v \in \ell^2\right\}$$

Remark. *Note that in the case where I is countably infinite, we often assume that $I = \mathbb{N}$, and ℓ^2 is the Hilbert space of square integrable sequences i.e. :*

$$\ell^2 := \left\{(u_i)_{i \in I} : \sum_{i \in I} u_i^2 < \infty\right\}$$

3.1.1 The Sobolev ellipsoid and the Gaussian sequence model

Definition 3.2 (Gaussian sequence model on the Sobolev ellipsoid)**.** *For $s, C > 0$, the* Sobolev ellipsoid *is $\mathcal{E}(s,C) := \left\{\theta \in \mathbb{R}^{\mathbb{N}} : \sum_{j=1}^\infty j^{2s}\theta_j^2 \le C\right\}$. Setting $\mu_\theta = \bigotimes_{j=1}^\infty \mathcal{N}(\theta_j, 1)$, the model class is:*

$$\mathcal{P}_G(s, C_G) := \{\mu_\theta \; ; \; \theta \in \mathcal{E}(s, C_G)\}.$$

Intuition. The weight j^{2s} in the Sobolev constraint penalizes high-frequency components: for the sum to stay bounded, the coefficients must satisfy $\theta_j = O(j^{-s})$. A large s forces fast decay (smooth signal, easy to detect), while a small s allows more energy at high frequencies (rough signal, harder to detect). This is directly reflected in Theorem 5.1: the exponent $\frac{4s+1}{2s}$ is decreasing in s.

Connection to the white noise model. This model arises naturally from the Gaussian white noise model $dY_t = f(t)\,dt + dW_t$. Projecting onto an orthonormal basis $\{\varphi_j\}$ of $L^2[0,1]$ gives $y_j = \theta_j + \varepsilon_j$ with $\varepsilon_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ and $\theta_j = \langle f, \varphi_j\rangle$. The Sobolev constraint on θ then corresponds exactly to a Sobolev regularity condition on f. Testing $H_0 : P_X = P_0$ is thus equivalent to testing whether an unknown signal f equals a given reference f_0.

Truncation. Since $\theta \in \mathcal{E}(s,C)$ forces $\theta_j \to 0$, only the first $J \asymp \varepsilon^{-1/s}$ coordinates carry a detectable signal; beyond index J the components are swamped by noise. This truncation is the key structural insight behind the separating set constructed in what follows.

3.2 Classical theory of minimax distinguishability

3.2.1 Asymptotic Framework

In this chapter, instead of considering only one hypothesis testing problem, we will follow an asymptotic approach in which we will be considering a family of Hypothesis testing problems indexed by a parameter $\varepsilon \to \varepsilon_0$. Depending on the setting, this parameter may be the dimensions $n \to \infty$, the sample size $N \to \infty$ or the white noise level $\varepsilon \to 0$. More generally, this corresponds to increasing the amount of available information.

Consider an observation $X_\varepsilon \in \mathcal{X}_\varepsilon$ generated by a probability measure $\mathbb{P}_{\varepsilon,\theta}$ on the measurable space $(\mathcal{X}_\varepsilon, \mathcal{A}_\varepsilon)$ where the parameter θ is unknown. Given two families of measurable subsets $\Theta_{\varepsilon,0}, \Theta_{\varepsilon,1}$, we will be testing the following hypotheses :

$$\mathcal{H}_0 : \theta \in \Theta_{\varepsilon,0} \ \text{ vs } \ \mathcal{H}_1 : \theta \in \Theta_{\varepsilon,1} \tag{3.2}$$

and we will be studying the performance of tests as $\varepsilon \to \varepsilon_0$. The statistical performance of a family of tests $\{\psi_\varepsilon\}$ can be studied using type I and II errors :

$$\alpha_\varepsilon(\psi_\varepsilon) = \sup_{\theta \in \Theta_{\varepsilon,0}} \mathbb{E}_{\varepsilon,\theta}\psi_\varepsilon \text{ and } \beta_\varepsilon(\psi_\varepsilon) = \sup_{\theta \in \Theta_{\varepsilon,1}} \mathbb{E}_{\varepsilon,\theta}[1 - \psi_\varepsilon]$$

Or :

$$\gamma_{\varepsilon,t}(\psi_\varepsilon) = t\alpha_\varepsilon(\psi_\varepsilon) + \beta_\varepsilon(\psi_\varepsilon) \ \text{ and } \ \gamma(t) = \inf_{\psi_\varepsilon \in \Psi}\Big(t\alpha_\varepsilon(\psi_\varepsilon) + \beta_\varepsilon(\psi_\varepsilon)\Big)$$

In the minimax setting, we say that a family of tests $\psi_\varepsilon^{(t)}$ is asymptotically minimax is :

$$\gamma_{\varepsilon,t}(\psi_\varepsilon^{(t)}) = \gamma_\varepsilon(t) + o(1) \ \text{ as } \ \varepsilon \to \varepsilon_0$$

Under the Neyman-Pearson setting, we consider families of tests ψ_ε satisfying $\alpha_\varepsilon(\psi_\varepsilon) \leq \alpha + o(1)$, and we say that a family $\psi_{\varepsilon,\alpha}$ is asymptotically minimax if :

$$\alpha_\varepsilon(\psi_{\varepsilon,\alpha}) \leq \alpha + o(1) \ \text{ and } \beta_\varepsilon(\psi_{\varepsilon,\alpha}) = \beta_\varepsilon(\alpha) + o(1) \ \text{ as } \ \varepsilon \to \varepsilon_0$$

We are interested in the study of the asymptotic properties of $\gamma_\varepsilon(t), \beta_\varepsilon(\alpha)$, thanks to Lemma 2.1, this can be done only by studying the asymptotic properties of $\gamma_\varepsilon := \gamma_\varepsilon(1)$. But this kind of problems is really difficult especially in the non parametric case, so we are interested in a simpler kind of problems called *distinguishability problem* (*rate asymptotic problem*) which can be summarized as follows :

- We want to obtain conditions for minimax distinguishability i.e. conditions for $\gamma_\varepsilon \to 0$, and construct *minimax consistent* families of tests ψ_ε such that $\gamma_\varepsilon(\psi_\varepsilon) \to 0$
- Obtain conditions for minimax nondistinguishability i.e. $\gamma_\varepsilon \to 1$

Proposition 3.1. *Let $\pi_{\varepsilon,0}, \pi_{\varepsilon,1}$ be families of priors such that :*

$$\pi_{\varepsilon,0}(\Theta_{\varepsilon,0}) \to 1 \ \textit{ and } \ \pi_{\varepsilon,1}(\Theta_{\varepsilon,1}) \to 1$$

Then, the following lower bounds hold true :

$$\begin{aligned} \beta_\varepsilon(\alpha) &\geq \beta_\varepsilon(\alpha; P_{\pi_{\varepsilon,0}}, P_{\pi_{\varepsilon,1}}) + o(1) \\ \gamma_\varepsilon &\geq \gamma_\varepsilon(P_{\pi_{\varepsilon,0}}, P_{\pi_{\varepsilon,1}}) + o(1) = 1 - \frac{1}{2}TV(P_{\pi_{\varepsilon,0}}, P_{\pi_{\varepsilon,1}}) + o(1) \end{aligned}$$

The problem that occurs often is that it is difficult to compute the total variation distance, so instead, we will use the χ^2-divergence that one can define as follows :

Definition 3.3. *Let P_0, P_1 be two probability measures on the same probability space. We define the χ^2-divergence between P_0 and P_1 as follows :*

$$|P_0 - P_1|_2^2 = \chi^2(P_0 \| P_1) = \begin{cases} \infty & \textit{if } P_0 \textit{ does not dominate } P_1 \\ \mathbb{E}_{P_0}\left[\frac{dP_1}{dP_0} - 1\right]^2 & \textit{otherwise} \end{cases} \tag{3.3}$$

Below, we will assume that $P_{\varepsilon,0}$ dominates all the other measures $P_{\varepsilon,\theta}$ for $\theta \in \Theta$ (This is fulfilled in the Gaussian sequence model)

Proposition 3.2 (χ^2 nondistinguishability)**.** *Ingster and Suslina [2012, Prop. 2.12]*

- *Let* $|P_{\varepsilon,1} - P_{\varepsilon,0}|_2 \to 0$*. Then* $TV(P_{\varepsilon,0}, P_{\varepsilon,1}) \to 0$ *and* $\gamma_\varepsilon(P_{\varepsilon,0}, P_{\varepsilon,1}) \to 0$*.*
- *If* $|P_{\varepsilon,1} - P_{\varepsilon,0}|_2 = \mathcal{O}(1)$*, then* $\limsup TV(P_{\varepsilon,0}, P_{\varepsilon,1}) < 2$ *and* $\liminf \gamma_\varepsilon(P_{\varepsilon,0}, P_{\varepsilon,1}) > 0$

Proof. The first statement follows from the inequalityc

$$|P_1 - P_0|_1 = E_{P_0}|L-1| \le \left(E_{P_0}|L-1|^2\right)^{1/2} = |P_1 - P_0|_2.$$

Let us establish the inequality

$$|P_1 - P_0|_2 \ge |P_1 - P_0|_1/2\left(1 - \tfrac{1}{2}|P_1 - P_0|_1\right)^{1/2}, \tag{3.4}$$

which yields the second statement. Let us consider the optimal test $\psi = \psi^{(1)} = \mathbf{1}_{\{L>1\}}$ for testing $H_0 : P = P_0$ against $H_1 : P = P_1$. By (**??**) we have

$$\gamma(\psi) = \gamma(P_0, P_1) = 1 - \tfrac{1}{2}|P_1 - P_0|_1 = 1 - E_{P_0}(L-1)\psi.$$

Then (3.4) follows from relations

$$\begin{aligned}
\tfrac{1}{2}|P_1 - P_0|_1 &= E_{P_0}(L-1)\psi \le \left(E_{P_0}|L-1|^2\right)^{1/2}\left(E_{P_0}\psi^2\right)^{1/2} \\
&= |P_1 - P_0|_2\, \alpha^{1/2}(\psi) \le |P_1 - P_0|_2\, \gamma^{1/2}(\psi) \\
&= |P_1 - P_0|_2\left(1 - \tfrac{1}{2}|P_1 - P_0|_1\right)^{1/2}.
\end{aligned}$$

□

3.2.2 Separation radius and goodness-of-fit

In Goodness-of-Fit problems (i.e. $\Theta_{\varepsilon,1} = \Theta^C_{\varepsilon,0}$ in Equation (3.2)), it is impossible to use directly minimax approach because :

$$\forall \varepsilon : \gamma_\varepsilon(\Theta_{\varepsilon,0}, \Theta_{\varepsilon,1}) \equiv 1$$

Therefore, we consider alternatives of the form :

$$\Theta_{\varepsilon,1} = \left\{\theta \in \Theta : d(\theta, \Theta_{\varepsilon,0}) \ge r_\varepsilon\right\}$$

Where d is a distance function and for $\theta \in \Theta : d(\theta, \Theta_{\varepsilon,0}) := \inf_{\theta_0 \in \Theta_{\varepsilon,0}} d(\theta, \theta_0)$. Now, our problem can be stated as follows : What are the radii r_ε in order to obtain minimax distinguishability $\gamma_\varepsilon \to 0$?
We will be studying this in the specific case of simple null-hypothesis of the form : $\mathcal{H}_0 : \theta \in \Theta_{\varepsilon,0} = \{\theta_0\}$.

Definition 3.4 (Critical radii)**.** *We say that a family r^*_ε is the critical radii if it satisfies the following dichotomy :*

$$\gamma_\varepsilon \to 0 \Leftrightarrow \frac{r_\varepsilon}{r^*_\varepsilon} \to \infty$$
$$\gamma_\varepsilon \to 1 \Leftrightarrow \frac{r_\varepsilon}{r^*_\varepsilon} \to 0$$

Moreover, we say that distinguishability conditions are in the classical form if the are in the form above.

TO DO

3.2.3 Minimax properties of χ^2 type tests

Now, return to the Gaussian model in the space of sequences ℓ^2, and condiser the following hypothesis problem :

$$\mathcal{H}_0 : v = 0 \ \text{ versus } \ \mathcal{H}_1 : v \in V_\varepsilon \subseteq \ell^2$$

Definition 3.5 (χ^2-type tests). *We say that a test ψ is of χ^2-type if there exists a family $w_\varepsilon = \{w_{\varepsilon,i}\}$ of sequences such that :*

$$\psi = \psi_{\varepsilon,w_\varepsilon;T} := \mathbb{1}_{t_\varepsilon > T_\varepsilon} \text{ where } t_\varepsilon = \sum_{i\in\mathbb{N}} w_{\varepsilon,i}(x_i-1)^2$$

In the context of the definition above, set :

$$h_\varepsilon(v) = \sum_{i\in\mathbb{N}} w_{\varepsilon,i}v_i^2;\ t_\varepsilon(v) = t_\varepsilon - h_\varepsilon(v);\ |w_\varepsilon| = \sup_{i\in\mathbb{N}} w_{\varepsilon,i}$$

Under the null hypothesis, one can note that $\mathbb{E}_0 t_\varepsilon$, moreover, one can assume the normalization $\sum w_{\varepsilon,i}^2 = \frac{1}{2}$ in order to ensure that $\mathrm{Var}(t_\varepsilon) = 1$

Lemma 3.1 (Minimax properties of χ^2-type tests). *Ingster and Suslina [2012, Lemma 3.1 & Corollary 3.3] Assume that the family w_ε satisfies $w_{\varepsilon,i} \geq 0$ as well as the normalization above. Then :*

$$Var_v(t_\varepsilon) \leq 1 + 4|w_\varepsilon|h_\varepsilon(v) \tag{3.5}$$

Moreover, if $h_\varepsilon := \inf_{v\in V_\varepsilon} h_\varepsilon(v, w_\varepsilon) \to \infty$ then for $T_\varepsilon = \frac{h_\varepsilon}{2}$, one has $\gamma(\psi_{w_\varepsilon,T_\varepsilon}, V_\varepsilon)$. Additionally, if one assumes that $|w_\varepsilon|$, then the statistics $t_\varepsilon(v)$ are asymptotically $\mathbb{N}(0,1)$ uniformly over $v \in \ell^2$ such that $h_\varepsilon(v) = o(|w_\varepsilon|^{-1})$, and the following inequalities hold true :

$$\beta(\psi_{\varepsilon,T_\alpha}, V_\varepsilon) \leq \Phi(T_\alpha - h_\varepsilon) + o(1),\ \gamma(\psi_{\varepsilon,T_\varepsilon}) \leq 2\Phi(-\frac{h_\varepsilon}{2}) + o(1)$$

Proof. The proof combines Lemma 3.1 of Ingster–Suslina (giving the variance bound and asymptotic normality of t_ε) with Corollary 3.3 (giving the minimax error bounds), the latter itself drawing on Corollaries 3.1–3.2.

Variance bound (??). Under P_v we have $X_i = \eta_i + v_i$ with $\eta_i \sim N(0,1)$ i.i.d. Writing $x = \eta + t$ with $\eta \sim N(0,1)$, $t \in \mathbb{R}$, a direct calculation gives

$$E(x^2) = 1 + t^2, \quad \mathrm{Var}(x^2) = 2 + 4t^2, \quad E\big((x^2 - Ex^2)^4\big) \leq c(1 + t^4), \tag{3.6}$$

where c is an absolute constant. Applying (3.6) coordinatewise and using the independence of the X_i,

$$E_v t_\varepsilon = \sum_i w_{\varepsilon,i}\, v_i^2 = h_\varepsilon(v), \qquad \mathrm{Var}_v(t_\varepsilon) = \sum_i w_{\varepsilon,i}^2\,(2 + 4v_i^2) = 1 + 4\sum_i w_{\varepsilon,i}^2\, v_i^2,$$

using the normalization $\sum_i w_{\varepsilon,i}^2 = 1/2$. Since $w_{\varepsilon,i}^2 v_i^2 \leq |w_\varepsilon|\, w_{\varepsilon,i} v_i^2$, we get

$$\mathrm{Var}_v(t_\varepsilon) \ \leq\ 1 + 4|w_\varepsilon| \sum_i w_{\varepsilon,i}\, v_i^2 \ =\ 1 + 4|w_\varepsilon|\, h_\varepsilon(v),$$

which is (??).

Asymptotic normality of $t_\varepsilon(v)$. Set $\xi_{\varepsilon,i} = w_{\varepsilon,i}(X_i^2 - 1)$, so that $t_\varepsilon = \sum_i \xi_{\varepsilon,i}$. By (3.6),

$$\sum_i E_v\big(\xi_{\varepsilon,i} - E_v\xi_{\varepsilon,i}\big)^4 \ \leq\ c\sum_i w_{\varepsilon,i}^4\,(1 + v_i^4).$$

Using $\sum_i w_{\varepsilon,i}^4 \leq |w_\varepsilon|^2 \sum_i w_{\varepsilon,i}^2 = |w_\varepsilon|^2/2$ and $\sum_i w_{\varepsilon,i}^4 v_i^4 \leq \big(\sum_i w_{\varepsilon,i}^2 v_i^2\big)^2 \leq |w_\varepsilon|^2 h_\varepsilon(v)^2$,

$$\sum_i E_v\big(\xi_{\varepsilon,i} - E_v\xi_{\varepsilon,i}\big)^4 \ \leq\ c\,|w_\varepsilon|^2\big(1 + h_\varepsilon(v)^2\big).$$

Since $\mathrm{Var}_v(t_\varepsilon) \geq 1$, the Lyapunov ratio is bounded by $c|w_\varepsilon|^2(1 + h_\varepsilon(v)^2)$. Under the hypotheses $|w_\varepsilon| = o(1)$ and $h_\varepsilon(v) = o(|w_\varepsilon|^{-1})$, this is $o(1)$ uniformly in the prescribed family, so the Lyapunov CLT yields

$$t_\varepsilon(v) = t_\varepsilon - h_\varepsilon(v) \ \xrightarrow{d}\ N\big(0,\, \mathrm{Var}_v(t_\varepsilon)\big) \ \xrightarrow{|w_\varepsilon|h_\varepsilon(v)\to 0}\ N(0,1),$$

uniformly over $v \in \ell^2$ such that $h_\varepsilon(v) = o(|w_\varepsilon|^{-1})$.

Type I error and the bound $\gamma(\psi_{w_\varepsilon,T_\varepsilon}, V_\varepsilon) \to 0$. Under H_0 (i.e. $v = 0$) we have $h_\varepsilon(0) = 0$, so Step 2 gives $t_\varepsilon \xrightarrow{d} N(0,1)$ under P_0. Hence $\alpha(\psi_{\varepsilon,T_\varepsilon}) = \Phi(-T_\varepsilon) + o(1)$ for any deterministic threshold T_ε.

For Type II error in the consistency regime, fix $v \in V_\varepsilon$, so $h_\varepsilon(v) \geq h_\varepsilon \to \infty$, and set $T_\varepsilon = h_\varepsilon/2$. By Chebyshev,

$$\beta(\psi_{w_\varepsilon,T_\varepsilon}, v) = P_v(t_\varepsilon \leq T_\varepsilon) \leq P_v\big(|t_\varepsilon - h_\varepsilon(v)| \geq h_\varepsilon(v) - h_\varepsilon/2\big) \leq \frac{\mathrm{Var}_v(t_\varepsilon)}{(h_\varepsilon(v) - h_\varepsilon/2)^2}.$$

Since $h_\varepsilon(v) \geq h_\varepsilon$ implies $h_\varepsilon(v) - h_\varepsilon/2 \geq h_\varepsilon/2$, and $\mathrm{Var}_v(t_\varepsilon) \leq 1 + 4|w_\varepsilon|h_\varepsilon(v)$,

$$\beta(\psi_{w_\varepsilon,T_\varepsilon}, v) \leq \frac{1 + 4|w_\varepsilon|h_\varepsilon(v)}{(h_\varepsilon/2)^2} = O\bigg(\frac{1}{h_\varepsilon^2} + \frac{|w_\varepsilon|h_\varepsilon(v)}{h_\varepsilon^2}\bigg) = o(1).$$

Taking the supremum over $v \in V_\varepsilon$,

$$\gamma(\psi_{w_\varepsilon,T_\varepsilon}, V_\varepsilon) = \alpha(\psi_{w_\varepsilon,T_\varepsilon}) + \sup_{v \in V_\varepsilon} \beta(\psi_{w_\varepsilon,T_\varepsilon}, v) \to 0.$$

Sharp bounds under $|w_\varepsilon| = o(1)$. Now assume additionally $|w_\varepsilon| = o(1)$. By Step 3, it suffices to consider $v \in V_\varepsilon$ with $h_\varepsilon(v) = O(1)$, since the case $h_\varepsilon(v) \to \infty$ yields $\beta \to 0$ already. For such v, Step 2 applies and gives $t_\varepsilon(v) \xrightarrow{d} N(0,1)$. Hence, with $T_\alpha = \Phi^{-1}(1-\alpha)$,

$$\begin{aligned}\beta(\psi_{\varepsilon,T_\alpha}, v) &= P_v\big(t_\varepsilon \leq T_\alpha\big) = P_v\big(t_\varepsilon - h_\varepsilon(v) \leq T_\alpha - h_\varepsilon(v)\big) \\ &= \Phi\big(T_\alpha - h_\varepsilon(v)\big) + o(1) \leq \Phi(T_\alpha - h_\varepsilon) + o(1),\end{aligned}$$

using the monotonicity of Φ and $h_\varepsilon(v) \geq h_\varepsilon$. Taking the supremum over $v \in V_\varepsilon$,

$$\beta(\psi_{\varepsilon,T_\alpha}, V_\varepsilon) \leq \Phi(T_\alpha - h_\varepsilon) + o(1).$$

For $T_\varepsilon = h_\varepsilon/2$, the Type I error is $\alpha(\psi_{\varepsilon,T_\varepsilon}) = \Phi(-h_\varepsilon/2) + o(1)$, while the Type II error at any $v \in V_\varepsilon$ with $h_\varepsilon(v) = O(1)$ satisfies

$$\beta(\psi_{\varepsilon,T_\varepsilon}, v) = \Phi\big(h_\varepsilon/2 - h_\varepsilon(v)\big) + o(1) \leq \Phi(-h_\varepsilon/2) + o(1).$$

Summing the Type I error and the supremum of the Type II error,

$$\gamma(\psi_{\varepsilon,T_\varepsilon}) \leq \Phi(-h_\varepsilon/2) + \Phi(-h_\varepsilon/2) + o(1) = 2\Phi(-h_\varepsilon/2) + o(1). \qquad \square$$

Chapter 4

Goodness-of-fit using Classifier-Accuracy tests

4.1 Setting of the problem

In this chapter, we study an ML-inspired approach to goodness-of-fit testing known as **Classifier-Accuracy Testing** (CAT), first explicitly described by Friedman Friedman [2004] and later studied in a rigorous minimax framework by Gerber et al. Gerber et al. [2023]. The central idea is to reduce a distributional testing problem to a classification problem: one trains a binary classifier on synthetic labeled samples, then uses its predictions on the actual data as a test statistic.

Definition 4.1 (Total variation distance)**.** *Let μ, ν be two probability measures on $(\mathcal{X}, \mathcal{A})$. Their* total variation distance *is:*

$$\mathrm{TV}(\mu, \nu) = \sup_{E \subset \mathcal{X}} |P(E) - Q(E)| = \frac{1}{2} \int_{\mathcal{X}} |f - g| \, d\lambda,$$

where the second equality holds when μ, ν admit densities f, g w.r.t. a common dominating measure λ.

Given a sample X of size $2n$ from an unknown P_X and a sample Y of size $2n$ from a known null P_Y, we wish to test $H_0 : P_X = P_Y$ versus $H_1 : \mathrm{TV}(P_X, P_Y) \geq \varepsilon$. A test $\psi : \mathcal{X} \to \{0, 1\}$ achieves error at most $\delta \in (0, \frac{1}{2})$ if $\max_i \max_{P \in H_i} \mathbb{P}(\psi(S) \neq i) \leq \delta$.

The CAT framework. Both samples are split as $X = X^{tr} \sqcup X^{te}$, $Y = Y^{tr} \sqcup Y^{te}$. A classifier $C : \mathcal{X} \to \{0, 1\}$ is trained on $D^{tr} = \{(X_i, 1)\} \cup \{(Y_i, 0)\}$, and represented through its **separating set** $S := C^{-1}(\{1\})$.

Definition 4.2 (Classifier-accuracy statistic)**.** *Given datasets $\{A_i\}_{i=1}^a$, $\{B_j\}_{j=1}^b$:*

$$T_S(A, B) := \frac{1}{a} \sum_{i=1}^{a} \mathbf{1}_{A_i \in S} - \frac{1}{b} \sum_{j=1}^{b} \mathbf{1}_{B_j \in S}.$$

The test rejects H_0 when $|T_S(X^{te}, Y^{te})|$ exceeds a threshold. A key finding of Gerber et al. [2023] is that restricting C to binary output does not sacrifice minimax optimality: what matters is not classification accuracy, but finding an *imbalanced* S with large separation $P_X(S) - P_Y(S)$ and small size $\tau(S) = \min\{P_X(S)P_X(S^c), P_Y(S)P_Y(S^c)\}$.

The main result of this chapter is the following lower bound on the sample complexity.

Theorem 4.1 ([Gerber et al., 2023, Table 1])**.** *Consider the goodness-of-fit problem $H_0 : P_X = P_0$ versus $H_1 : \mathrm{TV}(P_X, P_0) \geq \varepsilon$ with $P_0 \in \mathcal{P}_G(s, C)$. Then the minimax sample complexity satisfies:*

$$n(\varepsilon, \delta, s) \gtrsim \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} + \frac{\log(1/\delta)}{\varepsilon^2}.$$

The first term reflects the difficulty of nonparametric detection and depends on the smoothness s; the second is a universal information-theoretic floor for binary testing at confidence δ. The proof is developed in the following sections.

4.2 Preliminary results

The hard part of this kind of frameworks consists in finiding a good separating set $S \subset \mathcal{X}$, but we firstly need to define what "good" means.

Definition 4.3. *Let $S \subseteq \mathcal{X}$ be a measurable set, and μ, ν be two probability measures on $\mathcal{X}$. We define the separation and size of S respectively as follows :*

$$sep(S) = \mu(S) - \nu(S) \text{ and } \tau(S) = \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c))$$

Lemma 4.1. *Gerber et al. [2023, Lemma 1]*
Consider the hypothesis testing problem $\mathcal{H}_0 : p = q$ versus any other alternative $\mathcal{H}_1$. Suppose that the learner has constructed a separating set S such that :

$$\forall \mu, \nu \in \mathcal{H}_1 : |sep(S)| = |\mu(S) - \nu(S)| \geq \underline{sep} \text{ and } \forall \mu, \nu \in \mathcal{H}_0 \cup \mathcal{H}_1 : \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c)) \leq \bar{\tau}$$

Then, using only the knowledge of $\bar{\tau}$, the classifier-accuracy test with samples of size n from both p and q and an appropriate threshold achieves type-I and type-II errors at most δ provided that :

$$n \geq c \frac{\log(\frac{1}{\delta})}{\underline{sep}} \Big(1 + \frac{\bar{\tau}}{\underline{sep}}\Big)$$

for a large enough constant $c > 0$

Remark. *With Lemma 4.1, one can imagine that a separating can be referred to as "good" if $|sep(S)|$ is large under H_1, and $\tau(S)$ is small under both H_0 and $\mathcal{H}_1$ with probability $1 - \delta$*

In order to prove lemma 4.1, we will need a prior result on the Bernstein concentration of the classifier accuracy statistics.

Lemma 4.2. *Gerber et al. [2023, Lemma 8]*
Suppose $A_1, A_2, \ldots, A_n \overset{iid}{\sim}$ Ber(p) and $B_1, B_2, \ldots, B_m \overset{iid}{\sim}$ Ber(q). Let $\tau = \min(p(1-p), q(1-q))$, and let $\bar{A}, \bar{B}$ be their empirical means, respectively. There exists a universal constant $c > 0$ such that :

$$\mathbb{P}\Bigg(\left|\bar{A} - \bar{B}\right| \leq \frac{1}{2}\left|p - q\right| - \frac{1}{2}\sqrt{\frac{c\log(1/\delta)\tau}{\min(n,m)}} - \frac{1}{2}\frac{c\log(1/\delta)}{\min(n,m)} \Bigg) \leq \delta,$$
$$\mathbb{P}\Bigg(\left|\bar{A} - \bar{B}\right| \geq 2\left|p - q\right| + 2\sqrt{\frac{c\log(1/\delta)\tau}{\min(n,m)}} + 2\frac{c\log(1/\delta)}{\min(n,m)} \Bigg) \leq \delta$$

Proof of lemma 4.1. Using n test samples from (X, Y), consider the following classifier-accuracy test: we accept $\mathcal{H}_0$ if

$$\left| \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \leq \sqrt{\frac{c\bar{\tau}\log(1/\delta)}{n}} + \frac{c\log(1/\delta)}{n}$$

and reject the null hypothesis otherwise. Above, $c > 0$ is a large absolut constant, and note that the threshold only relies on the knowledge of $\bar{\tau}, n$ and δ
To analyse type-I and type-II errors , first, assume that $\mathcal{H}_0$ holds. Since sep$(S) = 0$ under the latter hypothesis, the second statement of Lemma 4.2 implies that we accept $\mathcal{H}_0$ with probability at least $1 - \frac{\delta}{2}$ if c is large enough. Moreover, if $\mathcal{H}_1$ holds, with probability at least $1-\frac{\delta}{2}$, the first statement of Lemma 4.2 implies that :

$$\left| \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \geq |\underline{\text{sep}}| + \Big(\sqrt{\frac{c\bar{\tau}\log(1/\delta)}{n}} + \frac{c\log(1/\delta)}{n}\Big)$$

By the assumed lower bound $n \geq c\frac{\log(\frac{1}{\delta})}{\underline{\text{sep}}}\Big(1 + \frac{\bar{\tau}}{\underline{\text{sep}}}\Big)$, we will reject $\mathcal{H}_0$ as desired. □

Proposition 4.1. *Gerber et al. [2023, Prop 1] Let $\mathbb{P}$ and $\mathbb{Q}$ be two probablity measures on a probability space $\mathcal{X}$, and let $\mathcal{C}$ be a possibly randomized classifier. Then the following inequality is true :*

$$TV(\mathbb{P}, \mathbb{Q}) \leq 2\sup\big\{\mathbb{P}(\mathcal{C}(X) = 0) - \mathbb{Q}(\mathcal{C}(X) = 0) \text{ over } X \text{ satisfying } \mathbb{P}(\mathcal{C}(X) = 0) = \mathbb{Q}(\mathcal{C}(X) = 1)\big\}$$

In the inequality above, the constant 2 is tight.

In order to prove Proposition 4.1 we will need the following lemma :

Lemma 4.3. *Let μ be a non-negative measure on a measurable space $\mathcal{X}$, and let $a, b : \mathcal{X} \to \mathbb{R}_+$ such that $\int a(x)d\mu(x) > 0$ and $\forall x \in \mathcal{X} : b(x) = 0 \implies a(x) = 0$. Then the following holds :*

$$\inf_{x \in supp(\mu)} \frac{a(x)}{b(x)} \leq \frac{\int a(x)d\mu(x)}{\int b(x)d\mu(x)} \leq \sup_{x \in supp(\mu)} \frac{a(x)}{b(x)}$$

Proof of Proposition 4.1. Let p, q be the densities of $\mathbb{P}, \mathbb{Q}$ with respect to a common dominated measure (e.g. $\mathbb{P} + \mathbb{Q}$), and define $E := \{p(x) > q(x)\}$, then $TV(\mathbb{P}, \mathbb{Q}) = \mathbb{P}(E) - \mathbb{Q}(E) \geq 0$. Without loss of generality, assume that $\mathbb{P}(E) + Q(\mathbb{E}) \geq 1$, and for $t \in [0, 1]$, define :

$$E_t := \left\{ . \frac{p(x) - q(x)}{p(x) + q(x)} \geq t \right\}$$

Then, the map $t \mapsto \mathbb{P}(E_t) + \mathbb{Q}(E_t)$ is non-increasing and left-continuous, and note that $E_0 = E$ and $E_1 = \emptyset$. Hence, the maximum $t^* := \max\{t \in [0, 1] : \mathbb{P}(E_t) + \mathbb{Q}(E_t) \geq 1\}$ exists. Now, define the randomized classifier $\mathcal{C}$ as follows :

$$\mathcal{C}(x) := \begin{cases} 0 & \text{if } x \in E_{(t^*)^+} \\ 1 & \text{if } x \notin E_{t^*} \\ \text{Ber}(x) & \text{if } x \in E_{t^*} \backslash E_{(t^*)^+} \end{cases}$$

Where :

$$E_{(t^*)^+} := \cap_{t > t^*} E_t \ \text{ and } \ r = \frac{1 - \mathbb{P}(E_{(t^*)^+}) - \mathbb{Q}(E_{(t^*)^+})}{\mathbb{P}(E_{t^*}) + \mathbb{Q}(E_{t^*}) - \mathbb{P}(E_{(t^*)^+}) - \mathbb{Q}(E_{(t^*)^+})} \in [0, 1]$$

Note that this classifier is balanced since $\mathbb{P}(\mathcal{C}(X) = 0) + \mathbb{Q}(\mathcal{C}(X) = 0) = 1$.
Now, for $t \in [0, 1]$, define :

$$f(t) := \begin{cases} \frac{\mathbb{P}(E_t) - \mathbb{Q}(E_t)}{\mathbb{P}(E_t) + \mathbb{Q}(E_t)} & \text{if } \mathbb{P}(E_t) + \mathbb{Q}(E_t) > 0 \\ 1 & \text{otherwise} \end{cases}$$

Now, let $0 \leq t \leq s \leq 1$, we want to show that $f(t) \leq f(s)$. Without loss of generality, assume that $f(s) < 1$ and that $\mathbb{P}(E_t) + \mathbb{Q}(E_t) > 0$. Notice also that (it can be proven using simple algebraic manipulations):

$$f(t) \leq f(s) \Leftrightarrow \frac{\int_{E_t \backslash E_s} p - q}{\int_{E_t \backslash E_s} p + q} \leq \frac{\int_{E_s} p - q}{\int_{E_s} p + q}$$

However, this inequality follows from Lemma 4.3. Thus, it holds that :

$$\frac{\mathbb{P}\big(\mathcal{C}(X) = 0\big) - \mathbb{Q}\big(\mathcal{C}(X) = 0\big)}{\mathbb{P}\big(\mathcal{C}(X) = 0\big) + \mathbb{Q}\big(\mathcal{C}(X) = 0\big)} \geq f(t^*) \geq f(0) = \frac{\mathbb{P}(E) - \mathbb{Q}(E)}{\mathbb{P}(E) + \mathbb{Q}(E)}$$

Plugging in $\mathbb{P}\big(\mathcal{C}(X) = 0\big) + \mathbb{Q}\big(\mathcal{C}(X) = 0\big) = 1$ and $\mathbb{P}(E) + \mathbb{Q}(E) \leq 2$ yields the final result. For the tightness, consider $p(x) = \mathbb{1}_{[0,1], q(x) = (1+\varepsilon)\mathbb{1}_{[0, \frac{1}{1+\varepsilon}]}, \mathcal{C}(x) = \mathbb{1}_{(\frac{1}{2+\varepsilon}, 1]}}$ and make $\varepsilon \to 0$. ☐

4.3 Performance of the CAT test in the Gaussian case

Suppose one have two samples X, Y of size $n \in \mathbb{N}$ from $\mu_{\theta^X} := \bigotimes_{j=1}^{\infty} \mathcal{N}(\theta_j^X, 1)$ and μ_{θ^Y}, respectively and such that μ^X, μ^Y have bounded Sobolev norm. Moreover, assume that $TV(\mu_{\theta^X}, \mu_{\theta^Y}) \geq \varepsilon > 0$, and let $\hat{\theta}^X, \hat{\theta}^Y$ denote the empirical means from the two samples respectively. Define the following statistic :

$$T(Z) = 2 \sum_{j=1}^{J} \left(\hat{\theta}_j^X - \hat{\theta}_j^Y \right) \left(Z_j - \frac{\hat{\theta}_j^X + \hat{\theta}_j^Y}{2} \right) \tag{4.1}$$

for some $J \in \mathbb{N}$ to be specified. We construct the separating set as follows :

$$\hat{S} := \left\{ Z \in \mathbb{R}^{\mathbb{N}} : T(Z) \geq 0 \right\} \tag{4.2}$$

The performance of this construction of separating set is summarized in the statement of the following result :

Proposition 4.2. *Gerber et al. [2023] There exists universal constants c, c' such that for $J = \lfloor c\varepsilon^{-\frac{1}{s}} \rfloor$ the following inequality holds provided $n \gtrsim \frac{1}{c'} n(\varepsilon, \delta)$:*

$$\mathbb{P}\Bigg(\mu_{\theta^X}(\hat{S}) - \mu_{\theta^Y}(\hat{S}) \geq c' \cdot \min\left(\sqrt{n\varepsilon^{\frac{1}{s}}}, \frac{1}{\varepsilon}\right) \cdot \varepsilon^2\Bigg) \geq 1 - \delta \tag{4.3}$$

In order to prove this proposition, we will need few lemmas and prior results.

Lemma 4.4 (Lipschitz concentration of Gaussians)**.** *Let Q be a d-dimensional standard gaussian, and let $f : \mathbb{R}^d \to \mathbb{R}$ be L-Lipschitz. Then $f(Q)$ is sub-Gaussian with variance proxy L^2*

Proof. See Appendix □

Lemma 4.5. *Let $a, b \in \mathbb{R}$, and let Z be a standard normal. Then :*

$$\mathbb{E}\Phi(aZ + b) = \Phi\left(\frac{b}{\sqrt{1 + a^2}}\right)$$

Proof. See Appendix □

Lemma 4.6. *Let sep$(\hat{S}) = \mu_{\theta^X}(\hat{S}) - \mu_{\theta^Y}(\hat{S})$ as defined above. There exists universal constants $c_i > 0$, i = 1,2,$\cdots$,5 such that for $J = \lfloor c_1\varepsilon^{-\frac{1}{s}} \rfloor$ and for $t \geq 0$, one has :*

$$\mathbb{E}[sep(\hat{S})] + \frac{c_2}{\sqrt{n}} \geq \frac{c_3\varepsilon^2}{\varepsilon + \sqrt{\frac{J}{n}}} \tag{4.4}$$

$$\mathbb{P}\left(\left|sep(\hat{S}) - \mathbb{E}sep(\hat{S})\right| \geq t + \frac{c_4}{\sqrt{n}}\right) \leq 2\exp(-c_5 n t^2) \tag{4.5}$$

Proof of Lemma 4.6. Let $\|\cdot\|, \langle\cdot,\cdot\rangle$ for the ℓ^2 norm/inner product restricted to the first J coordinates. Notice that given $\hat{\theta}^X, \hat{\theta}^Y, T(\theta)$ is a Gaussian random variable with :

$$\mathbb{E}T(\theta) = \|\hat{\theta}^Y - \theta\|^2 - \|\hat{\theta}^X - \theta\|^2 \text{ and } \mathrm{Var}(T) = 4\|\hat{\theta}^X - \hat{\theta}^Y\|^2$$

Now, define the random vectors U, V as follows :

$$U = \{\hat{\theta}^X_j - \hat{\theta}^Y_j\}_{j=1}^J$$
$$V = \{\hat{\theta}^X_j + \hat{\theta}^Y_j\}_{j=1}^J$$

One can easily see that these two vectors are independent, Gaussians with covariance matrix $\frac{2}{n}I_J$ and means equal to the first J coordinates of $\theta^X \mp \theta^Y$ respectively. Let Φ be the cumulative distribution function of the standard Gaussian and ϕ be its density. It can be shown that the separation can be written as :

$$\mathrm{sep}(\hat{S}) = f(\hat{\theta}^X) - f(\hat{\theta}^Y)$$

Where f is defined as follows :

$$f(\theta) = \Phi\left(\frac{\|\hat{\theta}^Y - \theta\|^2 - \|\hat{\theta}^X - \theta\|^2}{2\|\hat{\theta}^X - \hat{\theta}^Y\|^2}\right) = \Phi\left(-\frac{1}{2}\Big\langle V, \frac{U}{\|U\|}\Big\rangle + \Big\langle \theta, \frac{U}{\|U\|}\Big\rangle\right)$$

In order to make the dependences more clear, write $g(U, V) = f(\theta^X) - f(\theta^Y)$. Now, differentiating g and using the fact that ϕ is $\frac{1}{\sqrt{2\pi e}}$-Lipschitz yields :

$$\begin{aligned}
\|\nabla_V g(U, V)\| &= \Bigg\| -\frac{1}{2}\frac{U}{\|U\|}\Bigg(\phi\Big(-\frac{1}{2}\Big\langle V, \frac{U}{\|U\|}\Big\rangle + \Big\langle \theta^X, \frac{U}{\|U\|}\Big\rangle\Big) \\
&\qquad - \phi\Big(-\frac{1}{2}\Big\langle V, \frac{U}{\|U\|}\Big\rangle + \Big\langle \theta^Y, \frac{U}{\|U\|}\Big\rangle\Big)\Bigg)\Bigg\| \\
&\leq \frac{1}{\sqrt{8\pi e}}\left|\Big\langle \theta^X - \theta^Y, \frac{U}{\|U\|}\Big\rangle\right| \\
&\leq \frac{C}{\sqrt{8\pi e}}
\end{aligned}$$

By Lipschitz concentration of Gaussian random variable (Lemma 4.4), one can conclude that $g - \mathbb{E}[g|U]$ is sub-Gaussian with variance proxy $\frac{C^2}{4\pi en}$. Now, we study the concentration of $\mathbb{E}[g|U]$. First, note that :

$$-\frac{1}{2}\Big\langle V, \frac{U}{\|U\|}\Big\rangle + \Big\langle \theta, \frac{U}{\|U\|}\Big\rangle \Big| U \sim \mathcal{N}\Bigg(\Big\langle \theta - \frac{1}{2}(\theta^X + \theta^Y), \frac{U}{\|U\|}\Big\rangle, \frac{1}{2n}\Bigg)$$

Then, using the independence of U, V and lemma 4.5, we obtain :

$$\begin{aligned}\mathbb{E}[g(U,V)|U] &= \mathbb{E}[f(\theta^X) - f(\theta^Y)|U] \\ &= \Phi\Bigg(\frac{W}{\sqrt{4+\frac{2}{n}}}\Bigg) - \Phi\Bigg(-\frac{W}{\sqrt{4+\frac{2}{n}}}\Bigg)\end{aligned}$$

Where $W := \langle \theta^X - \theta^Y, \frac{U}{\|U\|}\rangle$. To simplify notations, let $\varphi := \Phi\Big(\frac{\cdot}{\sqrt{4+\frac{2}{n}}}\Big)$, by Lipschitness of Φ, we obtain that for $t \geq 0$:

$$\begin{aligned}\mathbb{P}\Big(\Big|\varphi(W) - \mathbb{E}\varphi(W)\Big| \geq t\Big) &\leq \mathbb{P}\Big(\Big|\varphi(W) - \varphi(\mathbb{E}W)\Big| \geq t - \|\varphi\|_{\text{Lip}}\sqrt{\text{Var}(W)}\Big) \\ &\leq \mathbb{P}\Big(|W - \mathbb{E}W| \geq \frac{t}{\|\varphi\|_{\text{Lip}}} - \sqrt{\text{Var}(W)}\Big)\end{aligned}$$

Moreover, one can get an analogous inequality for $-W$. Now, the last ingredient is to show that W concentrates well.

Lemma 4.7. *W is a sub-Gaussian random variable with variance proxy* $\frac{8}{n}$

We have already seen that :

$$\mathbb{E}[\text{sep}(\hat{S})] = \mathbb{E}[\varphi(W) - \varphi(-W)]$$

Again, by Lipschitzness and Lemma 4.5, we have that $|\mathbb{E}[\varphi(W)] - \varphi(\mathbb{E}W)| \lesssim \frac{1}{\sqrt{n}}$, hence :

$$\varphi(\mathbb{E}W) - \varphi(-\mathbb{E}W) \leq \mathbb{E}[\text{sep}(\hat{S})] + \Omega(\frac{1}{\sqrt{n}}) \tag{4.6}$$

Where the implied constant is universal. In order to simplify notations, let $\tau := \theta^X - \theta^Y, \sigma^2 = \frac{2}{n}$ and let Q be a $\mathcal{N}(0, I_J)$ random vector, then :

$$\mathbb{E}W = \mathbb{E}\Bigg[\Big\langle \tau, \frac{\tau + \sigma Q}{\|\tau + \sigma Q\|}\Big\rangle\Bigg] = \frac{1}{\sigma}\mathbb{E}[\langle \tau, \nabla_Q \|\tau + \sigma Q\|\rangle] = \frac{1}{\sigma}\mathbb{E}[\langle \tau, Q\rangle \|\tau + \sigma Q\|]$$

Where the last equality is given by Stein's identity. By rotational invariance of Gaussian distribution, we get :

$$\begin{aligned}\mathbb{E}W &= \frac{\|\tau\|}{\sigma}\mathbb{E}\Bigg[Q_1\sqrt{(\|\tau\| + \sigma Q_1)^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2}\Bigg] \\ &= \frac{\|\tau\|}{\sigma}\mathbb{E}\Bigg[Q_1\sqrt{(\|\tau\| + \sigma Q_1)^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2} - Q_1\sqrt{\|\tau\|^2 + \sigma^2 Q_1^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2}\Bigg] \\ &= 2\|\tau\|^2\mathbb{E}\Bigg[\frac{Q_1^2}{Q_1\sqrt{(\|\tau\| + \sigma Q_1)^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2} + Q_1\sqrt{\|\tau\|^2 + \sigma^2 Q_1^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2}}\Bigg]\end{aligned}$$

Now, by Cauchy-Schwartz inequality, and the fact that $\mathbb{E}|Q_1| = \frac{2}{\pi}$ is an absolute constant, we get that :

$$(\mathbb{E}|Q_1^2|)^2 \lesssim \mathbb{E}\Bigg[\frac{Q_1^2}{\sqrt{(\|\tau\| + \sigma Q_1)^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2} + \sqrt{\|\tau\|^2 + \sigma^2 Q_1^2 + \sigma^2 Q_2^2 + \cdots \sigma^2 Q_J^2}}\Bigg] \times (\|\tau\| + \sigma\sqrt{J})$$

Plugging everything into our expression for $\mathbb{E}W$, this yields :

$$\mathbb{E}W \gtrsim \frac{\|\tau\|^2}{\|\tau\| + \sigma\sqrt{J}}$$

To clarify notation, let us now write $\|\cdot\|_J$ for the ℓ_2-norm restricted to the first J coordinates. Taking $J = c\varepsilon^{-1/s}$ it holds that

$$\|\tau\|_J^2 \;=\; \|\tau\|^2 - \sum_{j>J} \tau_j^2 \;\geq\; \|\tau\|^2 - J^{-2s} \sum_{j>J} \tau_j^2 j^{2s} \;=\; \|\tau\|^2 - c^{-2s}\varepsilon^2 \|\tau\|_s^2.$$

Since $\|\tau\|_s \lesssim 1$ and $\|\tau\| \geq \varepsilon$ by assumption, we see that for large enough universal constant c we have $\|\tau\|_J \geq \varepsilon/2$. Since the map $x \mapsto x^2/(x+a)$ is increasing for $x, a > 0$, it follows that

$$\mathbb{E}W \;\gtrsim\; \frac{\varepsilon^2}{\varepsilon + \sqrt{J/n}}$$

for a universal implied constant (using $\sigma\sqrt{J} = \sqrt{2J/n} \asymp \sqrt{J/n}$). By the inequality $\Phi(x) - \Phi(-x) \geq x/2$ for $x \in [0,1]$ we obtain

$$\varphi(\mathbb{E}W) - \varphi(-\mathbb{E}W) \;\geq\; 1 \wedge \mathbb{E}W/2,$$

which, combined with Equation (4.6), completes the proof.

$\square$

Proof of Proposition 4.2. Apply Lemma 4.6 and set :

$$c = c_1 \;\text{ and }\; M := \frac{c_3\varepsilon^2}{\varepsilon + \sqrt{\frac{J}{n}}}$$

By Lemma 4.6, $\mathbb{E}[\text{sep}(\hat{S})] \geq M - \frac{c_2}{\sqrt{n}}$. Now, choose $t = \sqrt{\frac{\log(2/\delta)}{c_5 n}}$ so that $2\exp(-c_5 n t^2) = \delta$. By the tail bound given by Lemma 4.6 :

$$\mathbb{P}\Big(\text{sep}(\hat{S}) < \mathbb{E}[\text{sep}(\hat{S})] - t - \frac{c_4}{\sqrt{n}}\Big) \leq \delta$$

Hence, with probability at least $1-\delta$,

$$\text{sep}(\hat{S}) \geq M - \frac{c_2 + c_4}{\sqrt{n}} - \sqrt{\frac{\log(2/\delta)}{c_5 n}} \geq M - C\sqrt{\frac{\log(2/\delta)}{n}} \tag{4.7}$$

for some universal constant $C > 0$. Now, for $J = \lfloor c\varepsilon^{-\frac{1}{s}} \rfloor$, we have $\frac{J}{n} \asymp \frac{\varepsilon^{-\frac{1}{s}}}{n}$, hence :

$$\varepsilon + \sqrt{\frac{J}{n}} \asymp \begin{cases} \varepsilon & \text{if } \sqrt{n}\varepsilon^{1+\frac{1}{2s}} \geq 1 \\ \sqrt{\frac{J}{n}} & \text{,otherwise} \end{cases}$$

Using the asymptotics above, one can easily check that :

$$M \asymp \Big(\sqrt{n}\varepsilon^{\frac{1}{s}} \wedge \frac{1}{\varepsilon}\Big)\varepsilon^2$$

Also, the second term in Equation (4.7) is of order $\sqrt{\frac{\log(1/\delta)}{n}}$, under the condition $n \gtrsim \frac{1}{c'} n(\varepsilon, \delta, \mathcal{P}_G(s, C_G))$ (where $c' = \frac{c_3}{2}$ after absorbing all universal constants), this term is at most $\frac{M}{2}$. Therefore, with probability at least $1-\delta$:

$$\text{sep}(\hat{S}) \geq \frac{M}{2} \asymp \Big(\sqrt{n}\varepsilon^{\frac{1}{s}} \wedge \frac{1}{\varepsilon}\Big)\varepsilon^2$$

Which is the statement of Proposition 4.2. (Note that one can obtain the last inequality by adjusting the constants) $\square$

4.4 Sample complexity of CAT for Gaussian sequence model

Given two hypotheses H_0, H_1, our goal is to find a lower bound for the minimum achievable worst-case error. Firstly, we use the fact that for any random probability distributions P_0, P_1 with $\mathbb{P}(P_i \in H_i) = 1$ for $i = 1, 2$, one has that :

$$\min_{\psi} \max_{i=0,1} \sup_{P \in H_1} \mathbb{P}_{S\sim P}\big(\psi(S) \neq i\big) \geq \frac{1}{2}\big(1 - TV(\mathbb{E}P_0, \mathbb{E}P_1)\big) \tag{4.8}$$

Where $\mathbb{E}P$ denotes the mixture of P. Also, note that deriving a lower bound of order δ on the minimax error is equivalent to finding mixtures $\mathbb{E}P_0, \mathbb{E}P_1$ such that $1 - TV(\mathbb{E}P_0, \mathbb{E}P_1)$ is of order δ (is $\Omega(\delta)$)

Lemma 4.8. *Let μ, ν be two probability measures, then :*

$$1 - TV(\mu, \nu) \geq \frac{1}{2} e^{-KL(\mu \| \nu)} \geq \frac{1}{2\big(1 + \chi^2(\mu \| \nu)\big)}$$

Where $KL(\cdot \| \cdot)$ and $\chi^2(\cdot \| \cdot)$ denote the Kullback-Leibler and χ^2 divergences respectively.

Proof of Lemma 4.8. We will show that each one of the two inequalities hold true using Jensen inequality.

- Equivalently, we will show that :

$$\int \log\Big(\frac{d\mu}{d\nu}\Big) d\mu = KL(\mu \| \nu) \leq \log(1 + \chi^2(\mu \| \nu)) = \log\Big(\int \frac{d\mu}{d\nu} d\mu\Big)$$

  Which is a direct consequence of Jensen inequality (since $\log$ is concave)

- Instead of the inequality above, one will show that :

$$TV(\mu, \nu) \leq \sqrt{1 - e^{-KL(\mu \| \nu)}}$$

  First, notice that :

$$TV(\mu, \nu) = 1 - \int \big(d\mu \wedge d\nu\big) = \int \big(d\mu \vee d\nu\big) - 1$$

  This implies that :

$$\begin{aligned}
1 - TV(\mu, \nu)^2 &= \big(1 - TV(\mu, \nu)\big)\big(1 + TV(\mu, \nu)\big) \\
&= \int \big(d\mu \wedge d\nu\big) \int \big(d\mu \vee d\nu\big) \\
&\overset{\text{C-S}}{\geq} \left(\int \sqrt{fg}\right)^2 \quad \text{(Where } f, g \text{ are the densities)} \\
&= \exp\left(2 \log\Big(\int \sqrt{\frac{d\nu}{d\mu}} d\mu\Big)\right) \\
&= \exp\left(2 \log \mathbb{E}_\mu \left(\sqrt{\frac{d\mu}{d\nu}}\right)^{-1}\right) \\
&\geq \exp\left(-\mathbb{E}_\mu \log\left(\frac{d\mu}{d\nu}\right)\right) = \exp\Big(-KL(\mu \| \nu)\Big)
\end{aligned}$$

  And finally, one can conclude using the following inequality : $\sqrt{1 - x} \leq 1 - \frac{x}{2}$.

Combining the two points above yields the inequality. ☐

Now that we have all the necessary tools, we will prove the main theorem of this chapter. We will split this proof into many parts :

Proposition 4.3. *Gerber et al. [2023] There exists a constant K_0 such that for every ε small enough,*

$$n(\varepsilon, \delta, \mathcal{P}_G(s, C_G)) \geq K_0 \frac{log(1/\delta)}{\varepsilon^2} \tag{4.9}$$

Proof of Proposition 4.3. For a vector $\eta \in \{-1, 1\}^{\mathbb{N}}$, define the following probability measure :

$$\mathbb{P}_\eta := \bigotimes_{j=1}^{J} \mathcal{N}(\eta_j c_1 \varepsilon^{\frac{2s+1}{2s}}, 1) \otimes \bigotimes_{j > J} \mathcal{N}(0, 1) \tag{4.10}$$

Where $J = \lfloor c_2 \varepsilon^{-\frac{1}{s}} \rfloor$. And let $\eta_1, \eta_2, \cdots$ be iid uniform in $\{-1, 1\}$ and γ_η be the mean vector of $\mathbb{P}_\eta$. One can see that for any such η :

$$\|\gamma_\eta\|_s^2 = \sum_{j=1}^{\infty} j^{2s}\gamma_{\eta,j}^2 = \sum_{j=1}^{J} j^{2s}c^2\varepsilon^{\frac{2s+1}{s}} \leq c_1^2\varepsilon^{\frac{2s+1}{s}}\left(2c_2\varepsilon^{-\frac{1}{s}}\right)^{2s+1} \asymp c_1^2c_2^{2s+1}$$

$$\|\gamma_\eta\|^2 = \sum_{j=1}^{\infty}\gamma_{\eta,j}^2 \asymp c_1^2c_2\varepsilon^2$$

Then for $C_G > 0$ (Size of Sobolev's ellipsoid) fixed, one can choose c_1, c_2 independently of ε such that $\mathbb{P}_0, \mathbb{P}_\eta \in \mathcal{P}_G(s, C_G)$ almost surely and $\|\gamma_\eta\| = 10\varepsilon$. Then, for $\varepsilon \leq \frac{1}{10}$, one has :

$$TV(\mathbb{P}_0, \mathbb{P}_\eta) = 2\Phi\left(\frac{\|\gamma_\eta\|}{2}\right) - 1 \geq \varepsilon$$

Now, take $P_0 = \mathbb{P}_0^{\otimes 2n}, P_1 = \mathbb{P}_1^{\otimes n} \otimes \mathbb{P}_0^{\otimes n}$, then :

$$KL(P_0\|P_1) = nKL(\mathbb{P}_0\|\mathbb{P}_1) = nc_2\varepsilon^{-\frac{1}{s}}\frac{(c_1\varepsilon^{\frac{2s+1}{2s}})^2}{2} = Kn\varepsilon^2$$

Using Lemma 4.8 and Equation (4.8), we get the minimax error is at least :

$$\frac{1}{2}\big(1 - TV(P_0, P_1)\big) \geq \frac{1}{4}\exp(-KL(P_0\|P_1)) = \frac{1}{4}\exp(-Kn\varepsilon^2)$$

For this error to be less than δ, it is necessary to have :

$$\frac{1}{4}\frac{1}{4}\exp(-Kn\varepsilon^2) \leq \delta \Leftrightarrow n \geq \frac{\log(1/(4\delta))}{K\varepsilon^2}$$

Hence $n \gtrsim \frac{\log(1/\delta)}{\varepsilon^2}$ as claimed. ☐

Proposition 4.4. *Gerber et al. [2023] There exists a constant $K_1 > 0$ such that for every ε small enough :*

$$n(\varepsilon, \delta, \mathcal{P}_G(s, C_G)) \geq K_1\frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} \tag{4.11}$$

Proof. Let η be a sequence of iid uniform signs (i.e. with components from $\{-1, 1\}$ uniformly), and define the following probability measures :

$$P_0 = \mathbb{P}_0^{\otimes 2n}, P_1 = \mathbb{P}_0^{\otimes n} \otimes \mathbb{P}_\eta^{\otimes n}$$

By the independence of the components of η and their identical distribution, for $\omega = c_1\varepsilon^{\frac{2s+1}{2s}}$ the following mixture can be written as follows :

$$\mathbb{E}\mathbb{P}_\eta^{\otimes n} = \bigotimes_{j=1}^{J}\left(\frac{1}{2}\mathcal{N}(\omega, 1)^{\otimes n} + \frac{1}{2}\mathcal{N}(-\omega, 1)^{\otimes n}\right) \otimes \bigotimes_{j>J}\mathcal{N}(0, 1)^{\otimes n}$$

In order to simplify notations, define $M_n := \frac{1}{2}\mathcal{N}(\omega, 1)^{\otimes n} + \frac{1}{2}\mathcal{N}(-\omega, 1)^{\otimes n}$, hence :

$$KL(\mathbb{P}_0^{\otimes n}\|\mathbb{E}\mathbb{P}_\eta^{\otimes n}) = J \cdot KL(\mathcal{N}(0, 1)^{\otimes n}\|M_n) \tag{4.12}$$

Lemma 4.9. *For $\omega \in \mathbb{R}$ and $n \geq 1$:*

$$KL\left(\mathcal{N}(0, 1)^{\otimes n}\middle\|\frac{1}{2}\mathcal{N}(\omega, 1)^{\otimes n} + \frac{1}{2}\mathcal{N}(-\omega, 1)^{\otimes n}\right) \leq \frac{n^2\omega^4}{4} \tag{4.13}$$

Proof of Lemma 4.9. See Appendix ☐

By combining Equation (4.12) and Lemma 4.9, we obtain :

$$KL(\mathbb{P}_0^{\otimes n} \| \mathbb{E}\mathbb{P}_\eta^{\otimes n}) \leq J \cdot \frac{n^2 \omega^4}{4} = \frac{n^2}{4} \cdot J\omega^2 \cdot \omega^2$$

But $J\omega^2 = \|\gamma_\eta\|^2 = 100\varepsilon^2$ and $\omega^2 = c_1^2 \varepsilon^{\frac{2s+1}{s}}$ hence :

$$KL(\mathbb{P}_0^{\otimes n} \| \mathbb{E}\mathbb{P}_\eta^{\otimes n}) \leq \frac{100c_1^2}{4} n^2 \varepsilon^{\frac{4s+1}{s}} = \kappa \cdot n^2 \varepsilon^{\frac{4s+1}{s}}$$

Now, by combining Equation (4.8) and Lemma 4.8, in order to get an error less than δ, we need to have :

$$\kappa n^2 \varepsilon^{\frac{4s+1}{s}} \geq \log\left(\frac{1}{4\delta}\right)$$

Which yields $n \gtrsim \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}}$ □

Proof of Theorem 4.1. Using Proposition 4.3 and Proposition 4.4, we get that there exists two constants $K_0, K_1 > 0$ such that Equation (4.9) and Equation (4.11) hold true. Let $K = \min\{K_0, K_1\}$, one has :

$$\begin{aligned} n &\geq \max\left\{ K_0 \frac{\log(1/\delta)}{\varepsilon^2}, K_1 \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} \right\} \\ &\geq \max\left\{ K \frac{\log(1/\delta)}{\varepsilon^2}, K \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} \right\} \\ &\geq \frac{K}{2} \left(\frac{\log(1/\delta)}{\varepsilon^2} + \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} \right) \end{aligned}$$

□

Chapter 5

Conclusion

This project set out to understand when and why goodness-of-fit testing becomes hard in high-dimensional models, and how modern machine-learning-inspired procedures compare to classical statistical tests. The Gaussian sequence model over Sobolev ellipsoids provided a clean and mathematically tractable setting in which these questions could be made precise.

Chapter 3 revisited the classical theory of minimax distinguishability through the lens of χ^2-type tests. The key takeaway is that the smoothness of the signal class completely governs the difficulty of the problem: there is a sharp phase transition below which no test can do better than random guessing, and above which consistent testing becomes possible. This dichotomy is a fundamental feature of the problem, independent of the specific test used.

Chapter 4 then asked whether a classifier-based approach could achieve the same limits. The answer, perhaps surprisingly, is yes. We proved that any test requires at least

$$n(\varepsilon, \delta, s) \gtrsim \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} + \frac{\log(1/\delta)}{\varepsilon^2}$$

observations, and that the CAT test based on a truncated likelihood-ratio separating set matches this bound. The two terms in the lower bound tell different stories: the first captures the nonparametric hardness of the problem through the smoothness parameter s, while the second is simply the unavoidable cost of achieving high confidence δ in any binary testing problem.

Of course, CAT is not the only way to approach goodness-of-fit testing. Classical tests such as Kolmogorov–Smirnov, χ^2 goodness-of-fit, or kernel-based methods like the Maximum Mean Discrepancy have been studied for decades and remain widely used in practice. What makes CAT interesting is not that it outperforms these methods, but that it achieves the same fundamental limits while being naturally suited to the modern setting where distributions are only accessible through simulators.

More broadly, the results presented here suggest that the boundary between statistical theory and machine learning is less sharp than it might seem: a procedure motivated by classification can, with the right analysis, be shown to be optimal in a rigorous minimax sense.

Appendix

5.1 Some omitted proofs

Proof of Lemma 4.3. We will prove the LHS inequality, the RHS can be proven in the exact same way :

$$\int a(x)d\mu(x) = \int \frac{a(x)}{b(x)}b(x)d\mu(x)$$
$$\geq \Big(\inf_{x\in spt(\mu)} \frac{a(x)}{b(x)}\Big) \int b(x)d\mu(x)$$

The last inequality is implied by the fact that $\forall x \in \mathcal{X} : b(x) \geq 0$. □

Proof of Lemma 4.5. Let Z' be a standard Gaussian independent of Z, then :

$$\mathbb{E}\Phi(aZ+b) = \mathbb{P}(aZ+b \geq Z') = \mathbb{P}\Big(\frac{Z'-aZ}{\sqrt{1+a^2}} \leq \frac{b}{\sqrt{1+a^2}}\Big) = \Phi(\frac{b}{\sqrt{1+a^2}})$$

The last equality is given by the fact that $\frac{Z'-aZ}{\sqrt{1+a^2}} \sim \mathcal{N}(0,1)$. □

Proof of Lemma 4.7. Let $\tau := \theta^X - \theta^Y$ and $\sigma^2 \frac{1}{2n}$ and let Q be a standard Gaussian random vector such that :

$$W = \Big\langle \tau, \frac{\tau+\sigma Q}{\|\tau+\sigma Q\|}\Big\rangle \text{ in distribution}$$

Then, one can decompose W as follows :

$$\Big\langle \tau, \frac{\tau+\sigma Q}{\|\tau+\sigma Q\|}\Big\rangle = \Big\langle \frac{\tau}{\mathbb{E}(\|\tau\|)}, ss\Big\rangle$$

Correct the proof □

5.1.1 Proof of Lipschitz concentration for Gaussians

Firstly, note that we can restrict ourselves to smooth Lipschitz functions. Explicitely, if f is an L-Lipschitz function, one can consider its mollification $f_\varepsilon = f * \eta_\varepsilon$ where η is the standard mollifier. Moreover f_ε is smooth and converges uniformly on compact sets to f as $\varepsilon \to 0$. Now, we want to show the sub-Gaussianity of $f(Z)$ where $f : \mathbb{R}^d \to \mathbb{R}$ is an L-Lipschitz smooth function and Z is a standard Gaussian. This can be done using the Gaussian log-Sobolev inequality, which we will state in the following form :

Let γ_n denote the standard Gaussian measure on $\mathbb{R}^n$, that is, the law of $X \sim \mathrm{N}(0, I_n)$. We write $\mathbb{E}$ for expectation under γ_n and $\|\cdot\|_2$ for the Euclidean norm. For a nonnegative measurable function h with $\mathbb{E}[h] < \infty$ and $\mathbb{E}[h \log h] < \infty$, the *entropy of h relative to γ_n* is

$$\mathrm{Ent}_{\gamma_n}(h) := \mathbb{E}[h \log h] - \mathbb{E}[h] \log \mathbb{E}[h],$$

with the convention $0 \log 0 = 0$.

Lemma 5.1 (Gaussian logarithmic Sobolev inequality)**.** *For every smooth function $g \colon \mathbb{R}^n \to \mathbb{R}$ with $g, \nabla g \in L^2(\gamma_n)$,*

$$Ent_{\gamma_n}(g^2) \leq 2\,\mathbb{E}\big[\,\|\nabla g\|_2^2\,\big].$$

Lemma 5.2 (Herbst argument)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be smooth with $\|\nabla f(x)\|_2 \le L$ for every $x \in \mathbb{R}^n$. Then for every $\lambda \in \mathbb{R}$,*

$$\mathbb{E}\Big[e^{\lambda\,(f(X)-\mathbb{E}f(X))} \Big] \le \exp\!\left(\frac{\lambda^2 L^2}{2}\right).$$

Lemma 5.3 (Chernoff bound for sub-Gaussian variables)**.** *Let Y be a real-valued random variable satisfying $\mathbb{E}[e^{\lambda Y}] \le \exp\big(\lambda^2\sigma^2/2\big)$ for all $\lambda > 0$. Then for every $t > 0$,*

$$\mathbb{P}(Y \ge t) \le \exp\!\left(-\frac{t^2}{2\sigma^2}\right).$$

Theorem 5.1 (Gaussian Lipschitz concentration)**.** *Let $X \sim \mathbb{N}(0, I_n)$ and let $f\colon \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz. Then for every $t > 0$,*

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \ge t\big) \le \exp\!\left(-\frac{t^2}{2L^2}\right),$$

and consequently

$$\mathbb{P}\big(|f(X) - \mathbb{E}f(X)| \ge t\big) \le 2\exp\!\left(-\frac{t^2}{2L^2}\right).$$

The bound is independent of the dimension n.

Proof. By the smoothing reduction (i.e. the reduction to smooth Lipschitz functions) it suffices to prove the inequality under the additional assumption that f is smooth and satisfies $\|\nabla f(x)\|_2 \le L$ for every $x \in \mathbb{R}^n$, since the tail probability in the conclusion is upper semicontinuous with respect to the pointwise convergence $f_k \to f$.

Define the centered random variable

$$Y := f(X) - \mathbb{E}f(X),$$

so that $\mathbb{E}Y = 0$. The Herbst argument (Lemma 5.2), whose proof relies on the Gaussian logarithmic Sobolev inequality (Lemma 5.1), asserts that

$$\mathbb{E}\big[e^{\lambda Y} \big] \le \exp\!\left(\frac{\lambda^2 L^2}{2}\right) \qquad \text{for every } \lambda \in \mathbb{R}.$$

In other words, Y is sub-Gaussian with variance proxy $\sigma^2 = L^2$.

Applying the Chernoff bound (Lemma 5.3) with $\sigma^2 = L^2$ then yields, for every $t > 0$,

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \ge t\big) = \mathbb{P}(Y \ge t) \le \exp\!\left(-\frac{t^2}{2L^2}\right),$$

which establishes the upper-tail bound.

For the lower tail, observe that $-f$ is also L-Lipschitz, and that $\mathbb{E}(-f)(X) = -\mathbb{E}f(X)$. Applying the upper-tail bound just established to $-f$ in place of f gives

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \le -t\big) = \mathbb{P}\big((-f)(X) - \mathbb{E}(-f)(X) \ge t\big) \le \exp\!\left(-\frac{t^2}{2L^2}\right).$$

Combining the two tail bounds via a union bound,

$$\mathbb{P}\big(|f(X) - \mathbb{E}f(X)| \ge t\big) \le 2\exp\!\left(-\frac{t^2}{2L^2}\right).$$

None of the steps above involved any dependence on the dimension n, so the estimate is dimension-free. $\square$

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